

Rossmann Store Sales Prediction

End-to-End Time-Series Forecasting, LSTM Modeling & Streamlit Deployment

Author: Ankita Vaghela

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1. Executive Summary & Project Abstract

Accurate sales forecasting is vital for retail enterprises to optimize inventory management, streamline supply chain operations, and deploy effective staffing strategies. This project delivers an end-to-end deep learning framework designed to forecast daily sales across Rossmann drugstores. By leveraging historical daily store data, temporal feature engineering, and a Sequence LSTM neural network, the solution captures non-linear trends and multi-day seasonality. The final model is deployed via an interactive Streamlit dashboard allowing real-time revenue and customer traffic predictions.

Key Achievement: The Sequence LSTM model achieved an RMSPE score of **33.35%**, outperforming traditional regression baselines (Random Forest ~37.82%) by successfully modeling 7-day sequential time dependencies.

2. Problem Statement & Objectives

Rossmann operates over 3,000 drugstores across 7 European countries. Store managers face significant challenges in predicting daily sales volume due to complex factors including active promotional campaigns, state and school holidays, day-of-week traffic fluctuations, and varied store assortments.

- **Predictive Goal:** Build a robust machine learning/deep learning time-series model to forecast daily sales volume per store.
- **Customer Dynamics Rule:** Exclude the Customers column from direct feature training inputs (since actual customer count is unknown prior to the business day), while dynamically estimating footfall in the deployment dashboard.
- **Production Deployment:** Track experiment iterations using MLflow and host an intuitive Streamlit interface for non-technical retail managers.

3. Exploratory Data Analysis (EDA) & Insights

Comprehensive EDA was conducted to evaluate distribution patterns, temporal dynamics, and categorical feature impacts across historical store records.

3.1 Sales & Customer Distributions

- **Sales Clustering:** Daily store sales follow a right-skewed normal distribution, heavily clustered around **\$5,000 – \$7,000**. High-volume outlier stores peak near \$14,000.
- **Customer Footfall:** Daily footfall ranges primarily between **400 – 800 customers**, mirroring the sales distribution shape and proving a strong correlation between footfall and gross revenue.

- **Sales per Customer:** Consistently peaks around **\$10/customer**, confirming stable average basket sizes across standard operating days.

3.2 Categorical & Promotional Insights

- **Promotional Impact (Promo):** Active promotional campaigns yield a distinct lift in both overall revenue and daily customer volume. Promos are executed periodically rather than continuously.
- **Store Types & Assortments:** Store Type 'a' represents the largest share of records, whereas Store Type 'b' shows fewer entries but high sales density. Basic and Extended assortments generate over 90% of total revenue.
- **Holiday Dynamics:** Sales experience sharp demand spikes immediately preceding state holidays, followed by near-zero volume on official public holiday closures.

3.3 Temporal & Calendar Trends

- **Weekday vs. Weekend:** Average weekday sales (~\$7,000) significantly exceed weekend sales (~\$3,000), driven by regular business activities and Sunday store closures.
- **Quarterly Progression:** Sales show a steady upward trajectory from Q1 to Q4, reaching peak seasonal demand in Q4 during year-end holiday shopping.
- **Month-End Trends:** Higher average spend is recorded toward month-end (~\$6,500) compared to mid-month (~\$5,500), aligned with salary payout cycles.

4. Methodology & Model Architecture

Data preprocessing involved handling missing values, standardizing categorical encoding, and scaling feature sets using `StandardScaler`. To capture time-series sequential patterns, a multi-layer Long Short-Term Memory (LSTM) network was developed.

- **Sequence Window:** A rolling 7-day lookback window (`Time Steps = 7`) was implemented to feed historical temporal sequences into the LSTM layers.
- **Feature Scaling:** All input vectors (25 features) were normalized to ensure stable gradient descent convergence during neural network training.
- **Experiment Tracking (MLflow):** Evaluated multiple hyperparameter combinations, logging metrics, parameters, and saving model artifacts (`lstm_model.keras` and scalers).

5. Model Performance & Evaluation

Models were evaluated using Root Mean Square Percentage Error (RMSPE), the primary competition and business metric for Rossmann sales prediction.

Model Architecture	Evaluation Metric	Score	Performance Verdict
Random Forest Regressor	RMSPE	~37.82%	Baseline Solid tabular baseline, but lacks sequential memory.
Sequence LSTM (Time Steps = 7)	RMSPE	33.35%	Best Model Superior performance; effectively learns 7-day temporal trends.

6. Streamlit Deployment & Web Dashboard

An interactive web application (app.py) was developed to operationalize the trained LSTM model for business stakeholders.

- **User-Friendly Input UI:** Features intuitive selection dropdowns and Yes/No toggles for Promo, School Holiday, and State Holiday status.
- **Automated Feature Pipeline:** Automatically constructs and scales the 25-feature input array behind the scenes, preventing array shape mismatch issues.
- **Dual Outputs:** Displays real-time estimated Sales Revenue (\$) alongside dynamic Customer Footfall forecasts.
- **Reporting Capabilities:** Includes built-in visual bar charts and a 1-click CSV report export for store managers.

7. Strategic Business Recommendations

1. **Dynamic Inventory Allocation:** Utilize predicted high-revenue days (especially Q4 and promo weeks) to proactively scale inventory levels and prevent stockouts.
2. **Labor & Staff Optimization:** Schedule store shifts according to predicted daily customer footfall, ensuring optimal staff coverage during peak mid-week and pre-holiday periods.
3. **Targeted Promotion Planning:** Leverage the high responsiveness of Store Types 'a' and 'c' to run localized, high-impact promotional campaigns.